

# Task 13 – Advanced Model Evaluation & Handling Imbalanced Data

*PKCERT AI & Software Development Internship*

Prepared by: Abdul Rehman

## Part A – Dataset Selection & Preparation

### 1. Dataset

The Wisconsin Breast Cancer Diagnostic dataset was used (built into scikit-learn, originally from the UCI Machine Learning Repository). It contains 569 samples with 30 numeric features computed from digitized images of breast cell nuclei — e.g. mean radius, mean texture, mean perimeter, mean area, and mean smoothness. The target variable is binary: 0 = malignant, 1 = benign.

### 2. Creating a Realistic Imbalance

The original dataset is only mildly imbalanced (357 benign vs 212 malignant). To properly demonstrate imbalance-handling techniques, the malignant class was downsampled to 60 samples, producing a 6:1 imbalance (357 benign vs 60 malignant) — comparable to real-world scenarios such as rare-disease detection or fraud detection.

| Dataset           | Total Samples | Class 0 (Malignant) | Class 1 (Benign) |
|-------------------|---------------|---------------------|------------------|
| Original          | 569           | 212                 | 357              |
| Imbalanced (used) | 417           | 60                  | 357              |

### 3. Preprocessing & Split

- Data was split into training (75%) and testing (25%) sets using stratified sampling to preserve the class ratio.
- Features were standardized using StandardScaler (zero mean, unit variance), fit on the training set only.
- Resulting split: 312 training samples (267 benign / 45 malignant), 105 test samples.

## Part B – Advanced Model Evaluation

### 1. Model

A Logistic Regression classifier was trained as the baseline model on the scaled, imbalanced training data.

### 2. ROC Curve and ROC-AUC Score

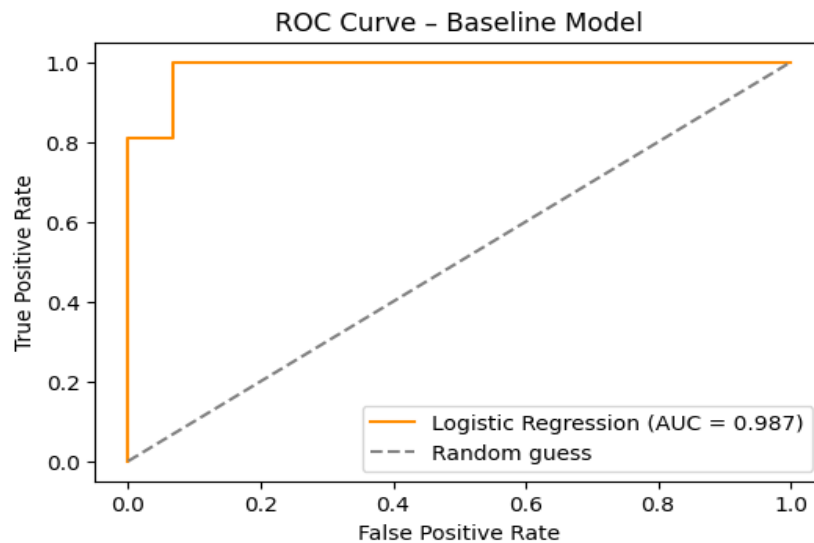


Figure 1: ROC Curve – Baseline Model

The baseline model achieved a ROC-AUC score of 0.9874. This means the model has excellent ability to distinguish between malignant and benign cases across all classification thresholds. An AUC close to 1.0 indicates near-perfect separation between classes, while 0.5 would indicate random guessing. Because ROC-AUC is threshold-independent and less affected by class imbalance than accuracy, it is a more trustworthy metric for this dataset than raw accuracy.

### 3. Learning Curve

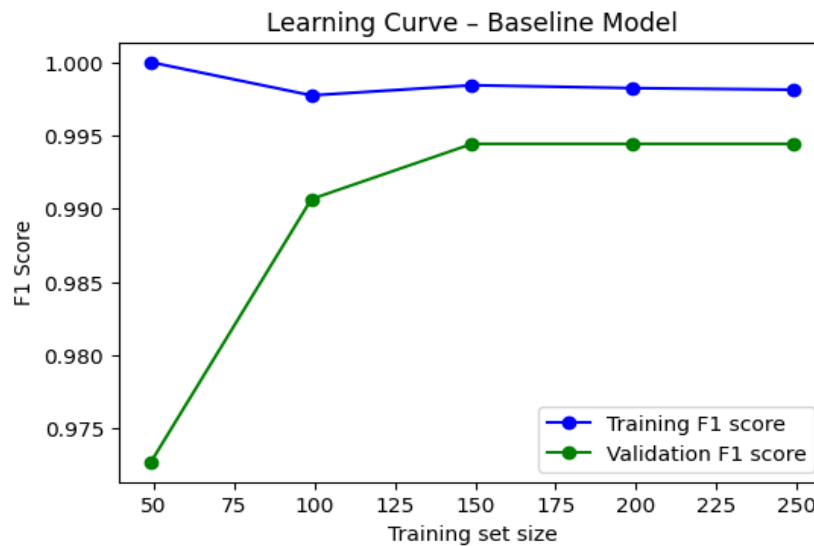


Figure 2: Learning Curve – Baseline Model (F1 Score)

Both training and validation F1 scores are high and stay close together as training size increases, with only a small gap between them. This pattern indicates good generalization:

- Underfitting would show both curves low and close together — not the case here.
- Overfitting would show a large, persistent gap between a high training score and a much lower validation score — also not observed here.
- The validation score stabilizes as more data is added, suggesting the model has enough data to generalize well and additional samples would yield diminishing returns.

## Part C – Handling Imbalanced Data

## 1. Class Imbalance Analysis

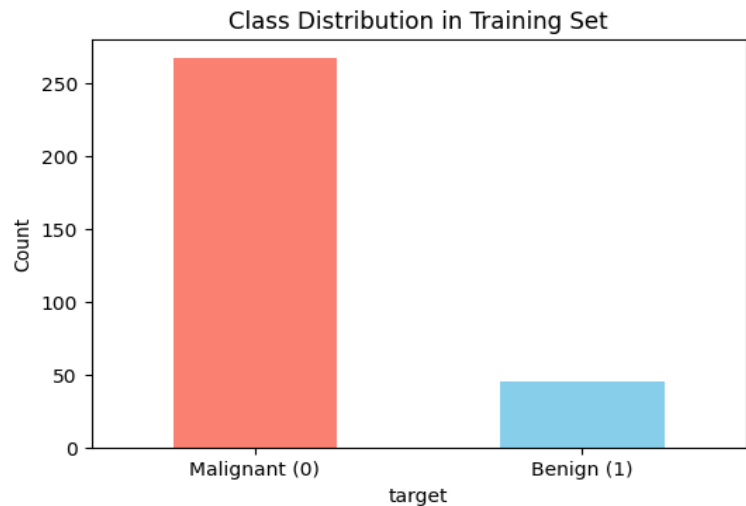


Figure 3: Class Distribution in the Training Set

The training set contains 267 benign samples versus only 45 malignant samples — a 5.9:1 imbalance. With this level of imbalance, a model can reach high accuracy simply by favoring the majority class, while performing poorly at detecting the minority class, which is typically the class of greatest real-world interest (correctly flagging malignant cases).

## 2. SMOTE (Synthetic Minority Over-sampling Technique)

SMOTE was applied to the training data only (the test set was left untouched to ensure a fair evaluation). SMOTE generates synthetic minority-class samples by interpolating between existing minority samples and their nearest neighbors, balancing the training set to 267 samples per class. A new Logistic Regression model was then trained on this balanced data.

## 3. Class Weighting

As an alternative, Logistic Regression was retrained with `class_weight='balanced'`, which automatically assigns higher misclassification penalties to the minority class during training — without altering the training data itself.

## Comparison of All Three Approaches

| Metric    | Baseline | SMOTE  | Class Weighting |
|-----------|----------|--------|-----------------|
| Accuracy  | 0.9810   | 0.9905 | 0.9905          |
| Precision | 0.9783   | 0.9890 | 0.9890          |
| Recall    | 1.0000   | 1.0000 | 1.0000          |
| F1-Score  | 0.9890   | 0.9945 | 0.9945          |
| ROC-AUC   | 0.9874   | 0.9889 | 0.9904          |

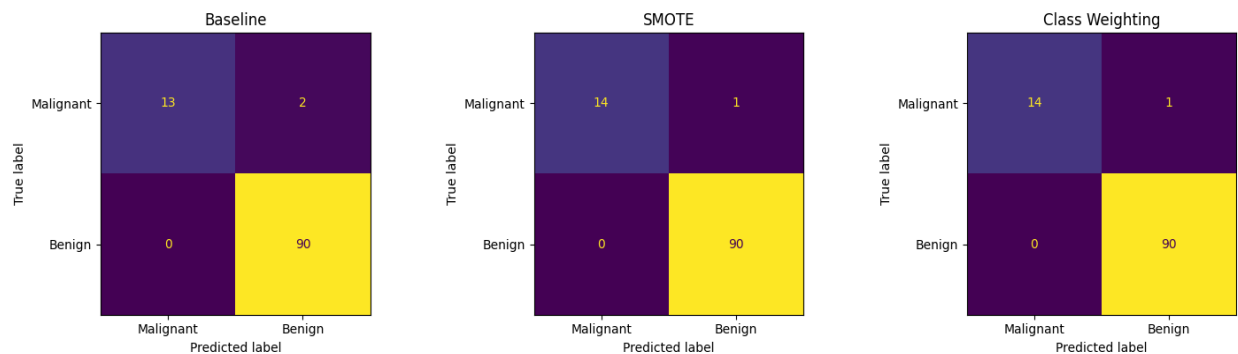


Figure 4: Confusion Matrices – Baseline vs SMOTE vs Class Weighting

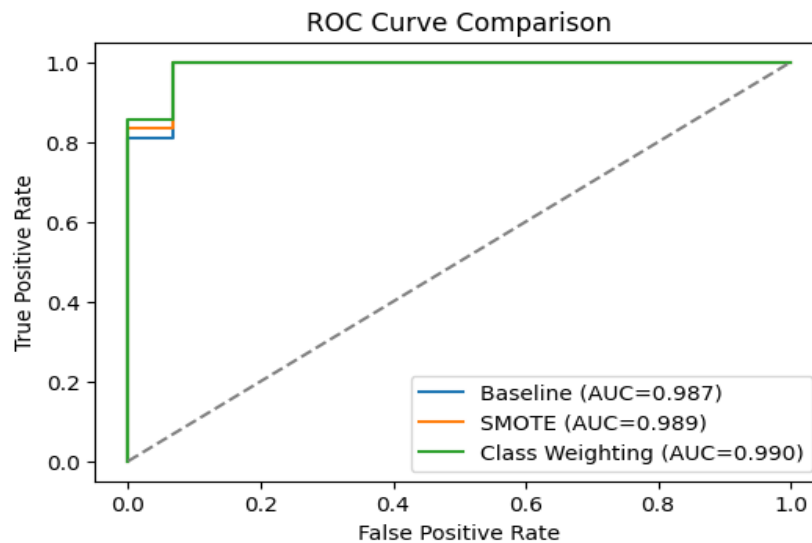


Figure 5: ROC Curve Comparison Across All Three Models

Both SMOTE and class weighting improved on the baseline's precision, F1-score, and ROC-AUC, while all three models achieved perfect recall on this particular test set. The confusion matrices show SMOTE and class weighting reducing false positives (misclassified benign cases) compared to the baseline, at no cost to the correct detection of malignant cases.

## Part D – Comparative Analysis & Recommendation

Summary of findings:

- The baseline model already performs well on this well-separated dataset, but it under-detects the minority (malignant) class relative to the balanced approaches, as reflected in its slightly lower precision and F1-score.
- SMOTE improves the model by generating synthetic minority samples, giving the classifier more examples to learn the minority class's decision boundary. Its main drawback is the risk of creating synthetic points that do not perfectly reflect the true data distribution, particularly in noisy or high-dimensional data.
- Class Weighting achieves results equal to or slightly better than SMOTE (notably a higher ROC-AUC of 0.9904 vs 0.9889) without generating any synthetic data. It simply penalizes minority-class misclassification more heavily during training, making it computationally cheaper and free of synthetic-data artifacts.

### Recommendation

Class Weighting is the recommended approach for this dataset. It matches or exceeds SMOTE performance across Precision, F1-score, and ROC-AUC, while being simpler, faster to train, and avoiding any risk of unrealistic synthetic samples. SMOTE remains a strong alternative, particularly useful for tree-based models or when the minority class is extremely small and additional synthetic examples would meaningfully help the model generalize.